

Predicting Rail Travel Duration using Machine Learning

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Presentation of thesis to obtain the Master of Science Degree
in Data Science and Engineering from Instituto Superior
Técnico

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June 2025

Welcome
Thank you to the jury



Research Motivation

Problem

Importance

Aims of this thesis

Abstract Summary

Predicting travel durations on regular services.

Proposed model offers a practical tool for improving service reliability and enhancing overall passenger experience in rail travel.

Data from Switzerland's FOT (Federal Office of Transport).

Keywords: Predictive Modeling, Regression Modeling, Rail Travel Prediction, Time Series Forecasting

Outline of Presentation

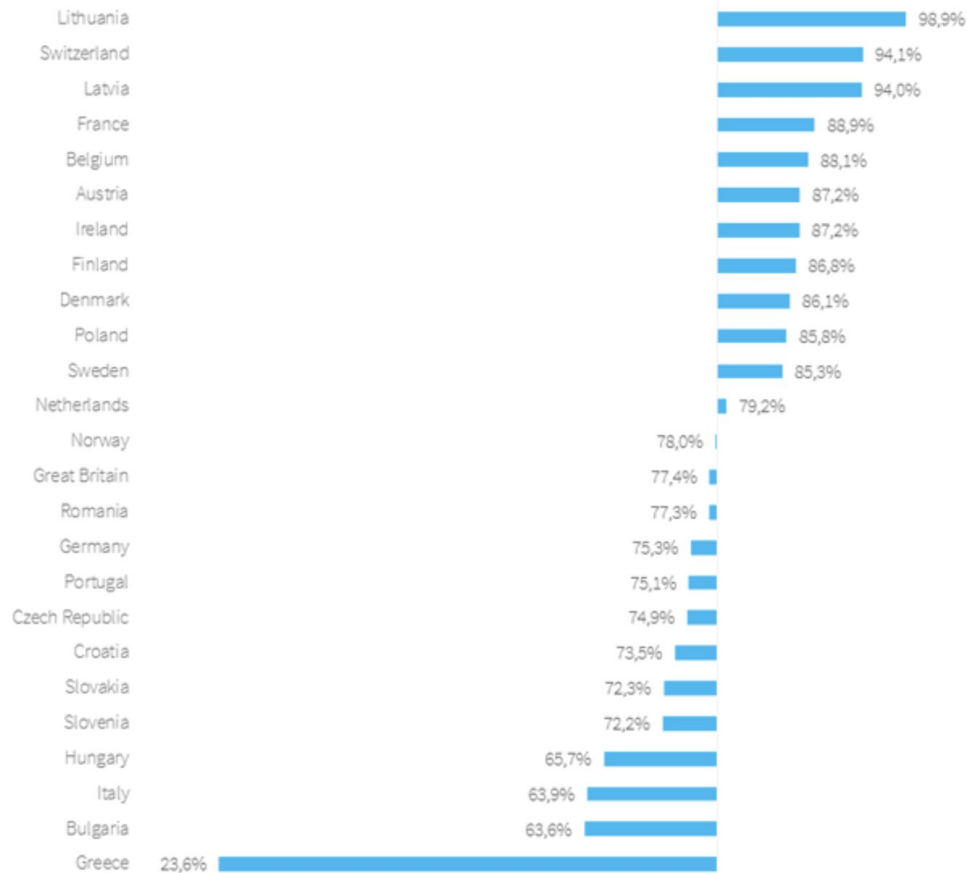
- Introduction to dataset
 - Data cleaning, descriptive statistics
- Learning models
 - Architecture
 - Evaluation metrics
- Measured outcomes
- Proposed model
- Limitations, extrapolations
- Conclusion

Current Standards

European Passengers' Federation
report from 2015-2016

Train punctuality within 5 minutes

Exception: Switzerland, punctuality
within 3 minutes (94.1%)



Current Standards

Vafaei and Yaghini, 2024

5 stations in Iran

ANN, RF, GB, XGB

Selected model: XGB

R^2 score of 0.9195

RMSE of 16 mins 51 sec

Prediction times and target variables in each of these times.

Prediction time	Train Position	Target variables
T_1	Before leaving Tehran Station	$AD_1, AA_2, AD_2, AA_3, AD_3, AA_4, AD_4, AA_5$
T_2	After leaving Tehran station	$AA_2, AD_2, AA_3, AD_3, AA_4, AD_4, AA_5$
T_3	Before leaving Garmsar station	$AD_2, AA_3, AD_3, AA_4, AD_4, AA_5$
T_4	After leaving Garmsar station	$AA_3, AD_3, AA_4, AD_4, AA_5$
T_5	Before leaving Semnan station	AD_3, AA_4, AD_4, AA_5
T_6	After leaving Semnan station	AA_4, AD_4, AA_5
T_7	Before leaving Shahrud station	AD_4, AA_5
T_8	After leaving Shahrud station	AA_5

Table 5 of Vafaei and Yaghini, 2024

Dataset Origin

Publicly available CSV files
uploaded daily by ODMCH of
all transportation (train, bus,
ship,

ODMCH works with SKI+,
FOT, RIF, NADIM

ODMCH (Open Data Platform
Mobility Switzerland)

SKI+ (System Tasks Customer
Information Plus)

FOT (Federal Office of Transport)

RIF (Rail Infrastructure Fund)

NADIM (National Data Networking
Infrastructure Mobility)

Dataset Origin

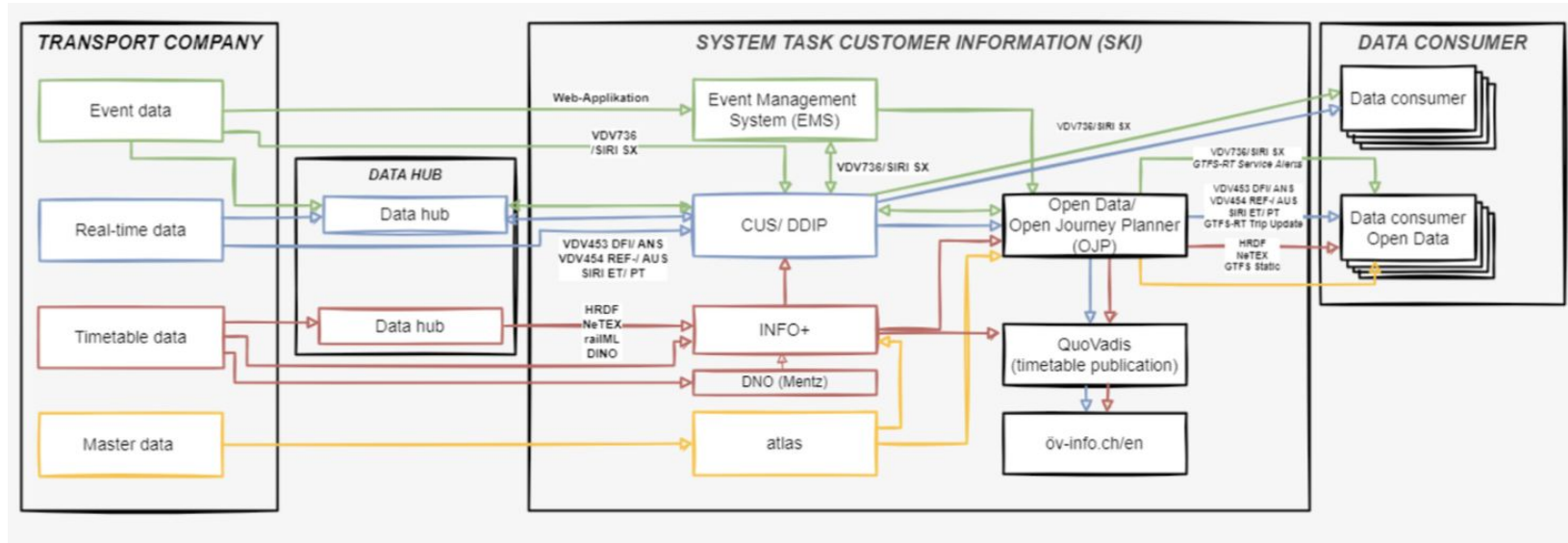


Image from ODMCH's Big Picture site

Data Cleaning

Products:

- Train
- Cog railway
- Metro
- Ship
- Tram
- Bus

Original field name (Swiss-German)	Translated field name (English)
BETRIEBSTAG	OPERATION DAY
FAHRT_BEZEICHNER	TRIP_IDENTIFIER
BETREIBER_ID	OPERATOR_ID
BETREIBER_ABK	OPERATOR_ABK
BETREIBER_NAME	OPERATOR_NAME
PRODUKT_ID	PRODUCT_ID
LINIEN_ID	LINE_ID
LINIEN_TEXT	LINES_TEXT
UMLAUF_ID	CIRCULATION_ID
VERKEHRSMITTEL_TEXT	TRANSPORTATION_TEXT
ZUSATZFAHRT_TF	ADDITIONAL_TRIP_TF
FAELLT_AUS_TF	FAILS_OFF_TF
BPUIC	BPUIC
HALTESTELLEN_NAME	STOP_NAME
ANKUNFTSZEIT	ARRIVAL TIME
AN_PROGNOSE	AN_FORECAST
AN_PROGNOSE_STATUS	AN_FORECAST_STATUS
ABFAHRTSZEIT	DEPARTURE TIME
AB_PROGNOSE_STATUS	AB_FORECAST_STATUS
DURCHFART_TF	TRANSIT_TF

Table 4.8 of this dissertation

Current Standards

Metric	Value (hh:mm:ss)
Arrival (Actual – Target) average	-00:00:47
Arrival (Actual – Target) median	00:00:39
Departure (Actual – Target) average	00:10:47
Departure (Actual – Target) median	00:01:06

Table 3.7 of this dissertation

Data Cleaning

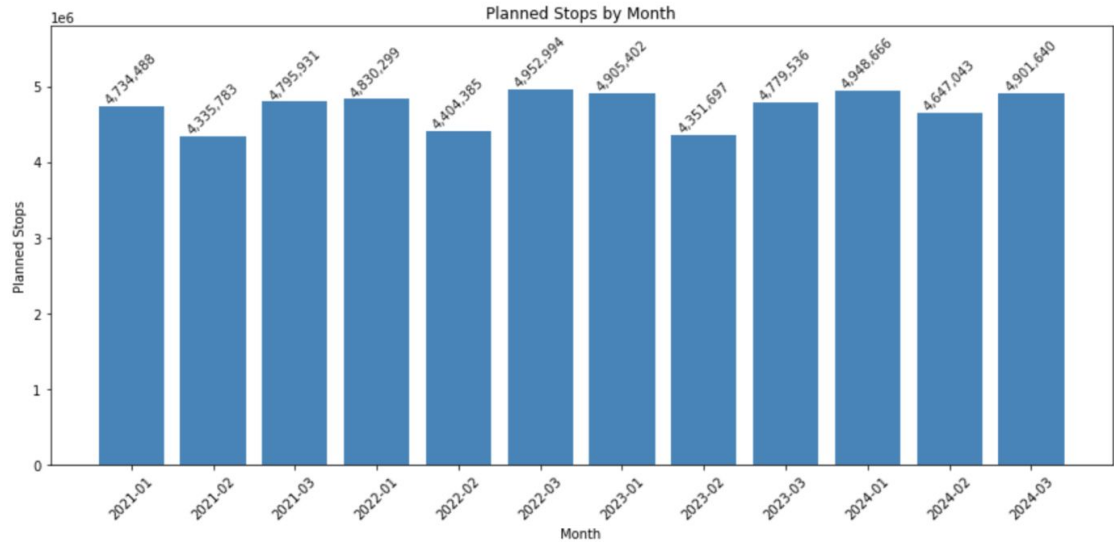


Figure 3.3 of this dissertation

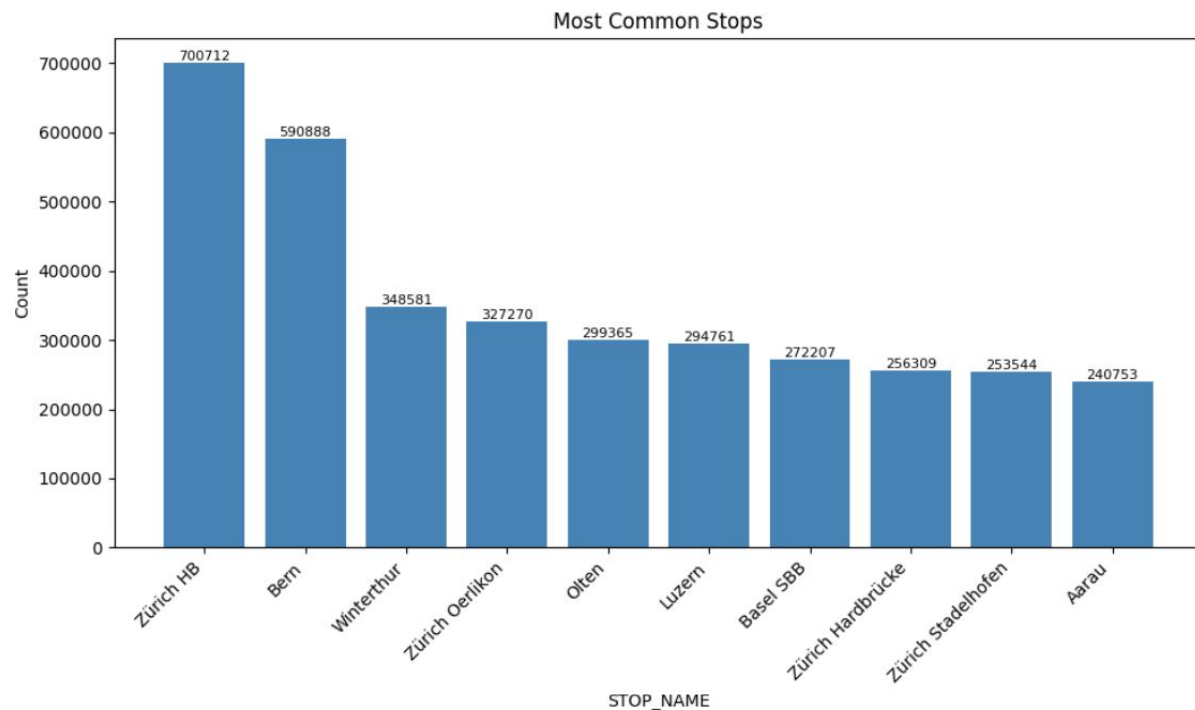
Years 2021, 2022, 2023, 2024

Months January, February, March

Dataset length 56,587,864 rows

22,967 unique lines

Data Cleaning



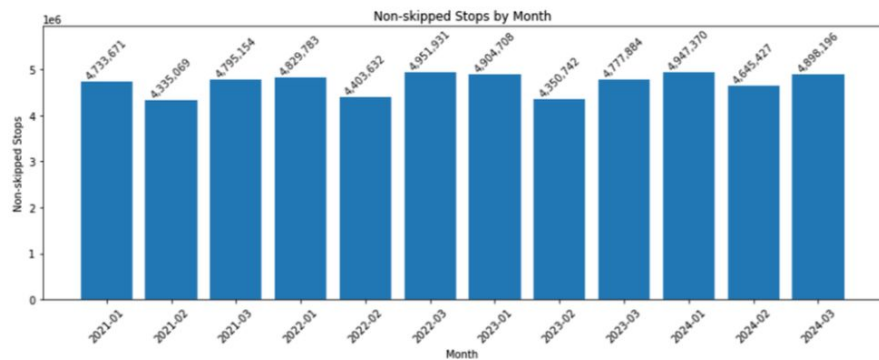
1,854 unique stop names

(“Aadorf” to “Zürich Wollishofen”)

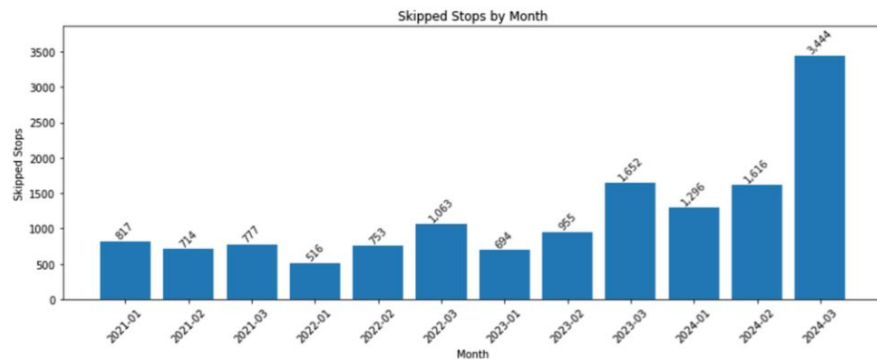
Figure 3.3 of this dissertation

Data Cleaning

Boolean column NO_STOP == False



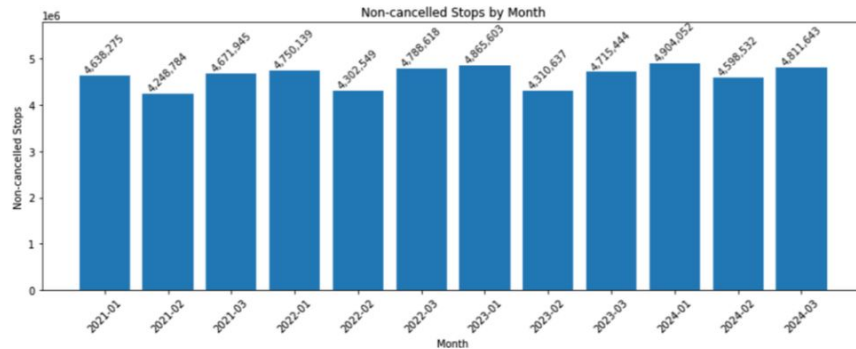
(A) Non-skipped stops



(B) Skipped stops

Data Cleaning

Boolean column CANCELLATION == False



(A) Non-cancelled stops



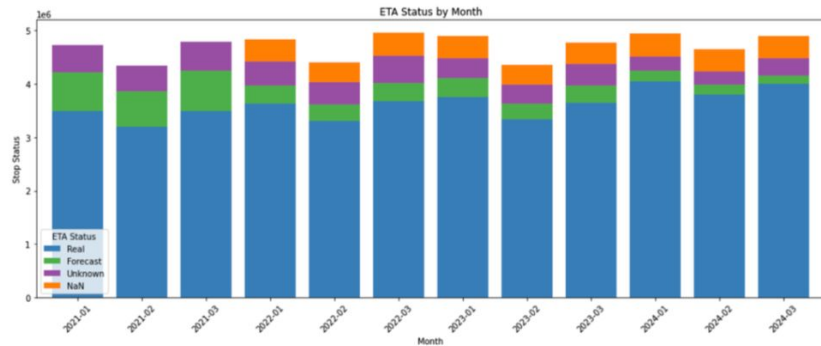
(B) Cancelled stops

Data Cleaning

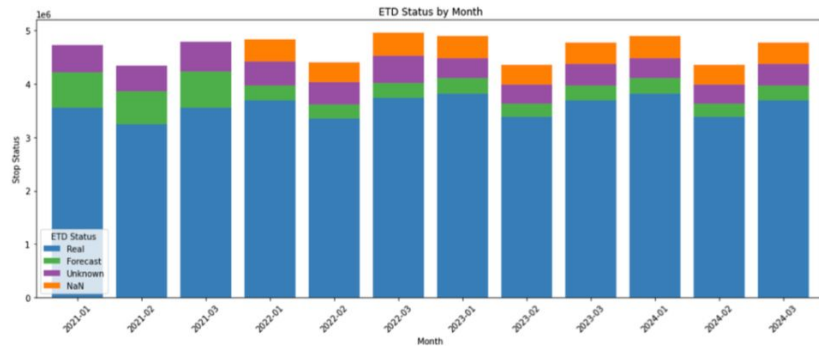
ETA_STATUS == Real and ETD_STATUS == Real

Relating to Actual Arrivals and Departures

Real, Forecast, Unknown, NaN



(A) ETA Status



(B) ETD Status

Data Cleaning

Shift function to create Previous Stop and Previous Departure time columns

First stop on a line missing Previous Stop and Previous Departure times

Last stop on a line missing Stop Name and Actual Departure times

Only kept rows where all existed:

Arrival, Departure, Stop Name, Previous Stop

Data Cleaning

Delta column

Actual Arrival - Previous Departure (seconds)

Only kept rows where $120 < \text{Delta} \leq 3600$

Expected journey length: 2 minutes to 1 hour

Data Cleaning

Dataset length 34,422,038 rows
(60.83% of uncleaned data length)

19,488 unique lines

1,480 unique stop names

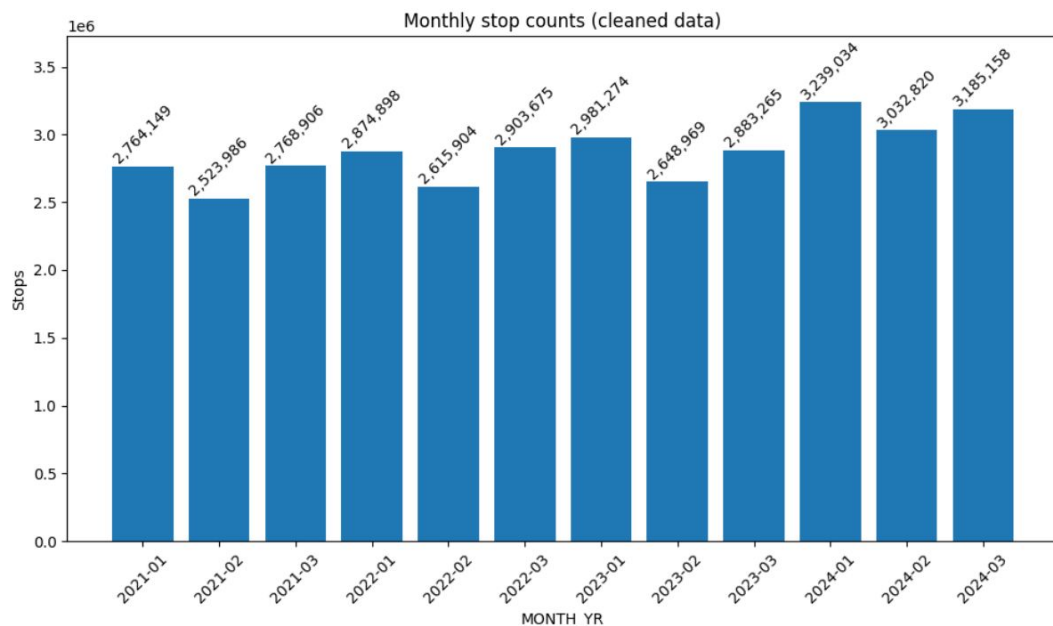


Figure 3.11 of this dissertation

Learning Models

Model	Selected?	Reason
Linear Regressor	Yes	Fast, helpful baseline
Decision Tree	Yes	Fast, can capture nonlinear relationships
Gradient Boosting	No	Does not optimize training as well as XGB
eXtreme Gradient Boosting (XGB)	Yes	Ensemble method with regularization, parallel computation
K-Nearest Neighbors	No	High computational and memory costs
Support Vector Regressor	No	High computational and memory costs

Feature and Target Variables

Features:

Departure Month, Day of Week, Hour, Minute

Stop Name Encoded, Previous Stop Name Encoded

Target:

Delta (seconds)

Architecture

80-20 Train-Test split

5-fold cross-validation

Random set seed for reproducibility

Randomized search (20 combinations)

Architecture

Randomized hyperparameter search

Hyperparameter	Values
max_depth	3, 5, 10, 15, None
min_samples_split	2, 5, 10
min_samples_leaf	1, 2, 5, 10
max_features	None, sqrt, log2

Table 3.8: Decision Tree search space

Hyperparameter	Values
n_estimators	100, 200, 300
max_depth	3, 5, 7, 10
learning_rate	0.01, 0.05, 0.1, 0.2
subsample	0.6, 0.8, 1.0
colsample_bytree	0.6, 0.8, 1.0

Table 3.9: XGB search space

Evaluation Metrics

R^2 score

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

$$SS_{\text{res}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

$$SS_{\text{tot}} = \sum_{i=1}^n (y_i - \bar{y})^2$$

RMSE (seconds)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Measured Outcomes

Linear Regressor

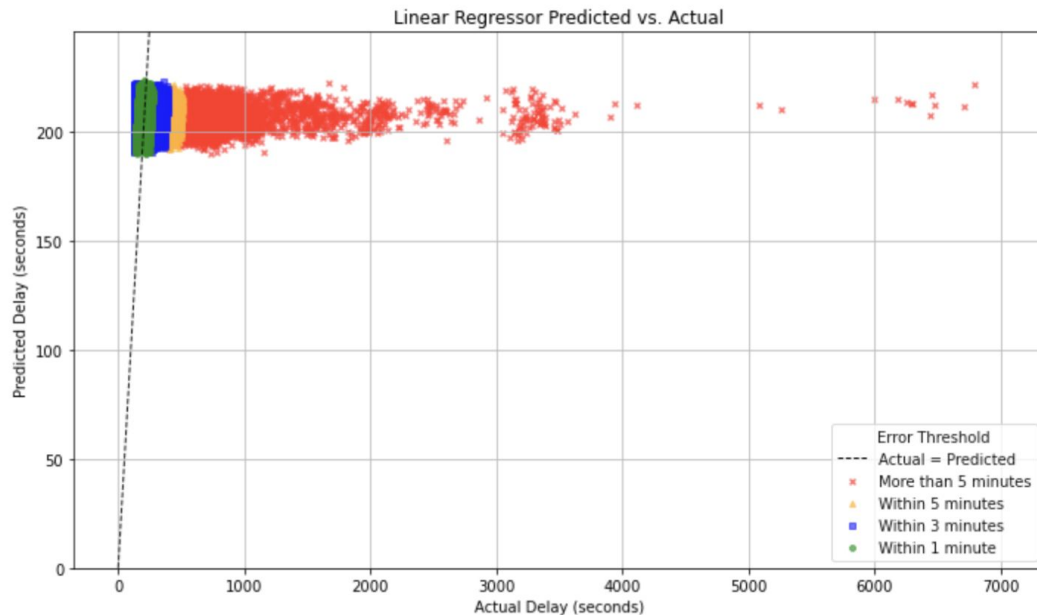


Figure 4.1

Metric	Value
Cross-validated R^2 Scores	[0.00036942, 0.00046652, 0.00048163, 0.00050388, 0.00050007]
Average Cross-validated R^2	0.0005
Test RMSE (seconds)	245.95

Table 4.1

Measured Outcomes

Decision Tree

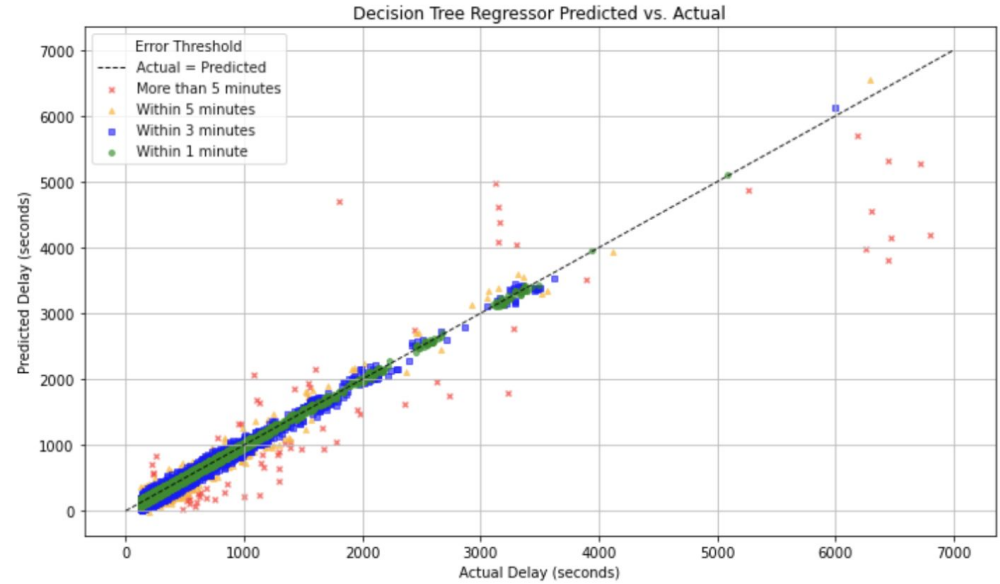


Figure 4.2

Metric	Value
Test R^2	0.9599
Test RMSE (seconds)	49.27

Table 4.3

Measured Outcomes

XGB

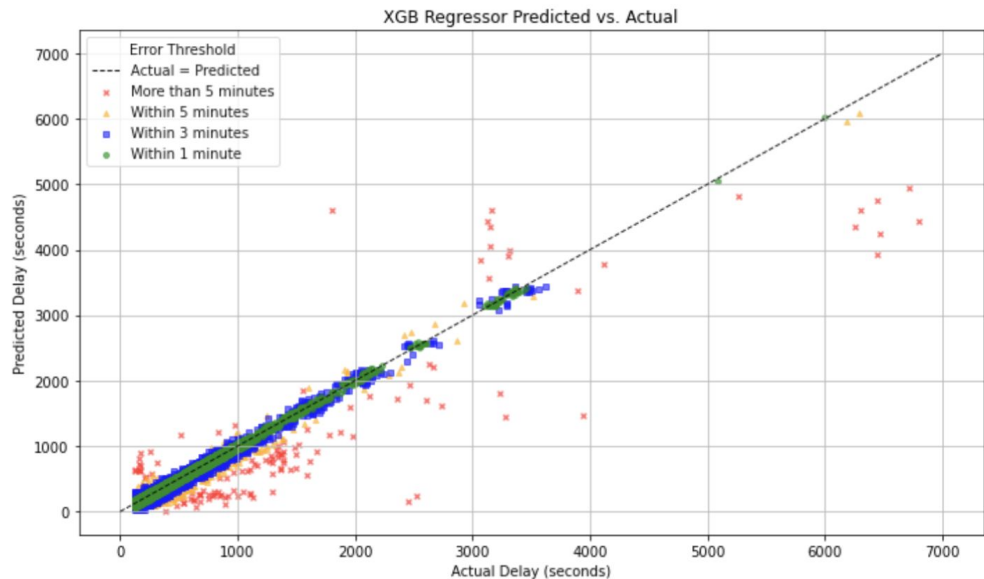


Figure 4.3

Metric	Value
Test R^2	0.9502
Test RMSE (seconds)	54.90

Table 4.5

Model Comparisons

Linear Regressor RMSE	245.95
Decision Tree RMSE	49.27
XGB RMSE	54.90

From Table 4.8

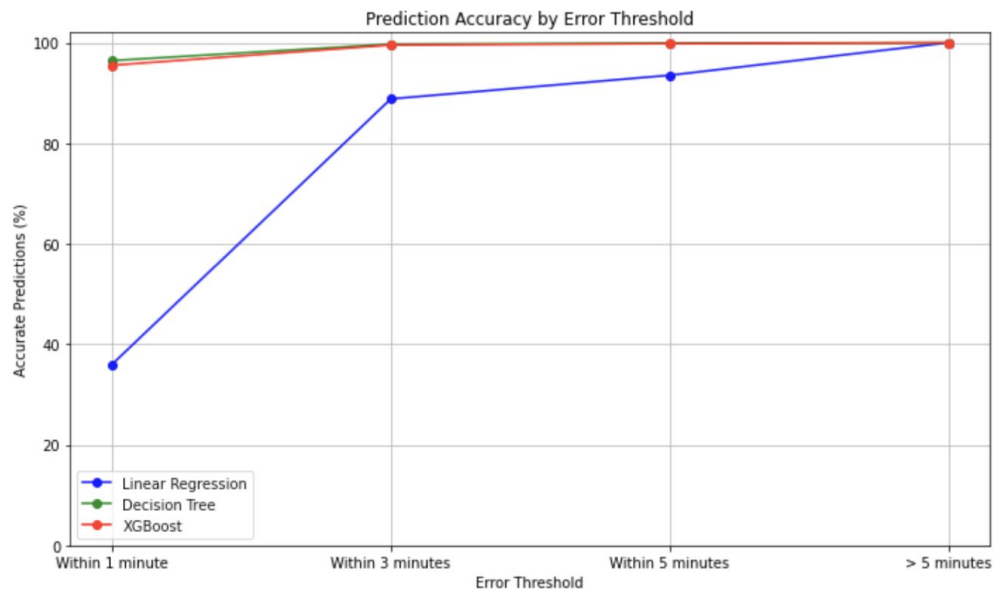


Figure 4.4

Category	Linear Regression		Decision Tree		XGB	
	Count	Cumul. %	Count	Cumul. %	Count	Cumul. %
Within 1 minute	2,478,019	35.995%	6,639,462	96.442%	6,575,190	95.508%
Within 3 minutes	3,637,094	88.826%	224,129	99.698%	277,704	99.542%
Within 5 minutes	322,772	93.514%	13,740	99.898%	18,129	99.805%
More than 5 minutes	446,523	100.000%	7,077	100.000%	13,385	100.000%

Table 4.7

Proposed Model

Decision Tree

Highest accuracy at each threshold

Highest R^2 score

Lowest RMSE

Parameter	Value
min_samples_split	10
min_samples_leaf	10
max_features	None
max_depth	None

Table 4.4

Current Standards Comparison

Proposed model accuracy within 3 minutes:
99.698%

Proposed model R^2 score: 0.9599

Proposed model RMSE: 49.27

European Passengers' Federation report
punctuality within 3 minutes: 94.1%

Vafaei and Yaghini, 2024

R^2 score of 0.9195

RMSE of 16 mins 51 sec

Uncleaned data metrics (From Table 4.8)

Metric	Value (seconds)
Arrival (Actual – Target) average	47
Arrival (Actual – Target) median	39
Departure (Actual – Target) average	647
Departure (Actual – Target) median	66

Limitations

Computation and memory resources

Selected months from dataset

One data set

Could be improved by weather, rail maintenance data

Extrapolations

Proposed model would likely suit similar data

- Same feature and target variables

- Similar times between stops

- Other countries

- Other transportation types

Conclusions

Publicly available data from ODMCH (Open Data Platform Mobility Switzerland) and FOT (Federal Office of Transport)

Data cleaning steps

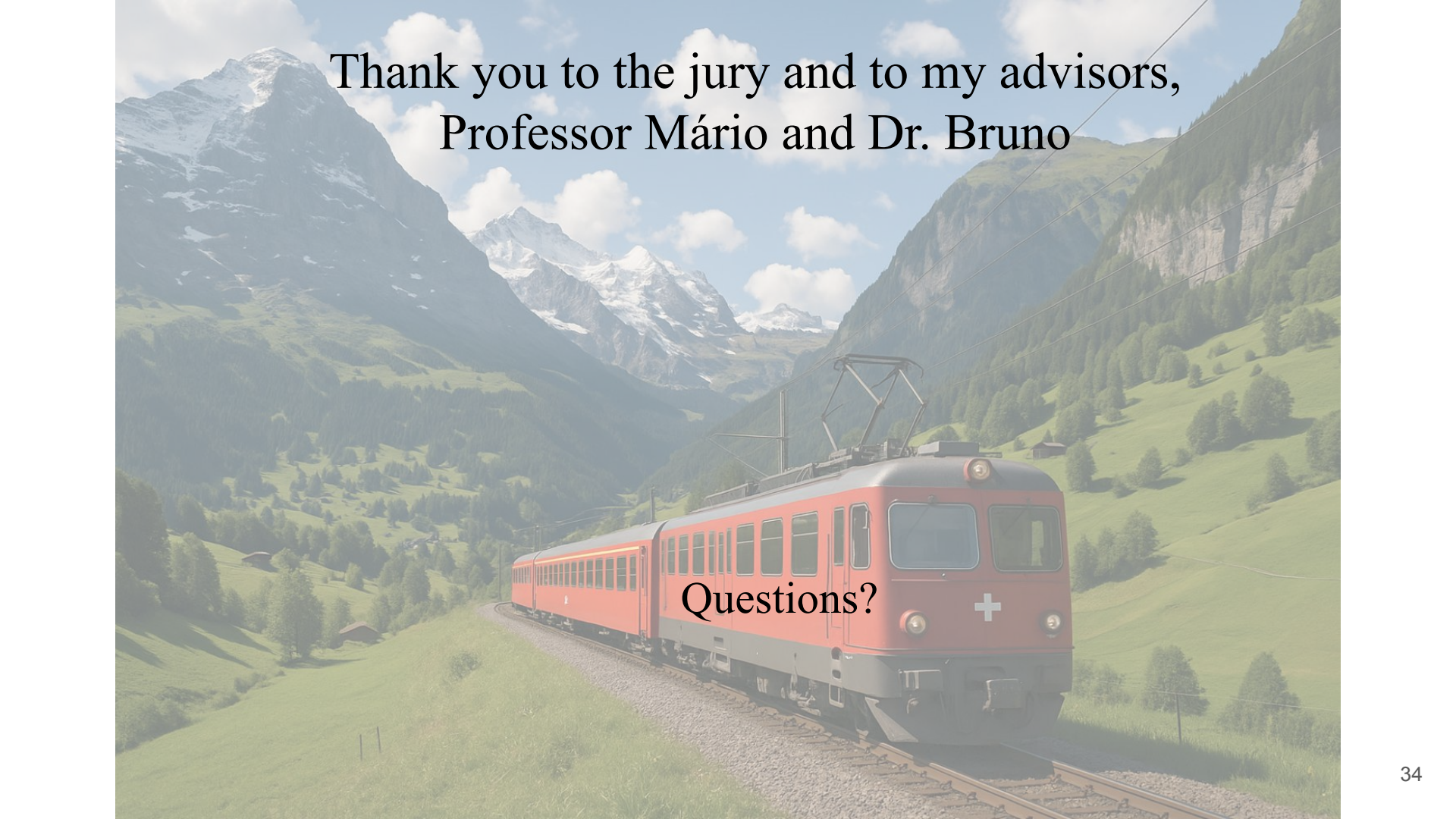
Column names, skipped stops, cancellations, Real status, constrained Delta

Architecture, evaluation metrics

Running models

Comparing evaluation metrics among models, current standards

Choosing model to propose



Thank you to the jury and to my advisors,
Professor Mário and Dr. Bruno

Questions?